# I. Correctness and Parameter Analysis

## A. Correctness

Let $\alpha = \frac{e^{1/5}}{\sqrt{2\pi}}$. Probabilities in this correctness analysis include the sampling of $(s, e)$ during key generation as well as preprocessing. We use the standard subgaussian norm bound for the ideal discrete-Gaussian samples and apply it directly to independent sums; no convolution is replaced by an exact spherical discrete Gaussian. Concretely, for independent centered samples $x_k$ with subgaussian parameters $\sigma_k$, the coefficientwise moment-generating-function bound and independence give subgaussian parameter $(\sum_k \sigma_k^2)^{1/2}$ for $\sum_k x_k$; the standard vector-norm tail bound with $\alpha = \frac{e^{1/5}}{\sqrt{2\pi}}$. and $N \geq 16\lambda$ is at most $2^{-\lambda}$. In particular, for $a = |\mathsf{act}| \leq a_{\max}$, except with probability at most $6 \cdot 2^{-\lambda}$, all six bounds

$$\left\|\sum_j y_j\right\|_2 \leq \alpha\sqrt{aN}\,\sigma_2, \qquad \left\|\sum_j e'_j\right\|_2 \leq \alpha\sqrt{aN}\,\sigma_2,$$
$$\left\|\sum_j r_{j,1}\right\|_2, \left\|\sum_j r_{j,2}\right\|_2 \leq \alpha\sqrt{aN}\,\sigma_r,$$
$$\|s\|_2, \|e\|_2 \leq \alpha\sqrt{N}\,\sigma_1$$

hold simultaneously when $N \geq 16\lambda$. The indices in these sums range over $\mathsf{act}$. Since $\|c\|_1 = \tau$, Young's inequality for negacyclic convolution gives

$$\|sc\|_2 \leq \tau\|s\|_2, \qquad \|ec\|_2 \leq \tau\|e\|_2.$$

This avoids treating $sc$ or $ec$ as an independent spherical Gaussian.

To support every active set of size at most $a_{\max}$, define the conservative bounds

$$\begin{aligned}
\beta_z(a_{\max}) &= \alpha\sqrt{N}\,(\sqrt{a_{\max}}\,\sigma_2 + \tau\sigma_1)\,,\\
\beta_r(a_{\max}) &= \alpha\sqrt{a_{\max}N}\,\sigma_r,\\
\beta_{\widetilde{e}}(a_{\max}) &= \alpha\sqrt{N}\,(\sqrt{a_{\max}}(\sigma_2+\sigma_r) + \tau\sigma_1) + \frac{q\tau}{2q_1}\sqrt{N},\\
\beta_\Delta(a_{\max}) &= \frac{q_2}{q}\beta_{\widetilde{e}}(a_{\max}) + \sqrt{N}.
\end{aligned}$$

The verifier uses $\beta_z = \beta_z(a_{\max})$, $\beta_r = \beta_r(a_{\max})$, and $\beta_\Delta = \beta_\Delta(a_{\max})$.

For an honest execution, pairwise-mask cancellation and Shamir reconstruction give

$$\begin{aligned}
z &= \sum_{j\in\mathsf{act}} (y_j + \lambda_{\mathsf{act},j}s_jc + m_j)\\
&= \sum_{j\in\mathsf{act}} y_j + sc, \qquad r = \sum_{j\in\mathsf{act}} r_{j,2}.
\end{aligned}$$

Let

$$\eta_1 = \widehat{b} - \mathsf{Lift}_1(b), \qquad \|\eta_1\|_\infty \leq \frac{q}{2q_1},$$

and define the residual, using canonical centered representatives in $\mathcal{M}_q$, by

$$\widetilde{e} = \sum_{j\in\mathsf{act}} (e'_j + r_{j,1}) + ec - \eta_1 c.$$

For $X = az + h_{\mathsf{sid}}r - \mathsf{Lift}_1(bc_{\mathbb{Z}})$, where the product is formed in $R_{q_1}$ before lifting, the honest aggregate token satisfies

$$C_{\mathsf{act}} = \lfloor X + \widetilde{e}\rceil_{p_2}\,.$$

The bounds above and $\|\eta_1 c\|_2 \leq (q\tau/(2q_1))\sqrt{N}$ imply $\|\widetilde{e}\|_2 \leq \beta_{\widetilde{e}}(a_{\max})$. The modulus-switching difference bound then gives

$$\|\Delta\|_2 \leq \frac{q_2}{q}\|\widetilde{e}\|_2 + \sqrt{N} \leq \beta_\Delta(a_{\max}).$$

Finally, the definition of $\Delta$ yields

$$C' = \Delta + \lfloor X\rceil_{p_2} = C_{\mathsf{act}},$$

so the verifier recomputes the honest challenge. In this construction, at least one honest signer belongs to every admissible active set because $|\mathcal{C}| < t \leq |\mathsf{act}|$. A successful honest signing execution has therefore bound the exact context before its response collection completes. The persistent binding read matches the context reconstructed above. Hence the added check introduces no further cryptographic correctness error with the ideal available service. An implementation's probability of an unavailable or invalid read must be accounted for separately; correctness does not promise progress after adversarial burns that prevent a signing execution from completing.

A union bound over the six displayed subgaussian events gives

$$\varepsilon^{\mathsf{pt}}_{\mathsf{corr}}(\lambda) \leq 6 \cdot 2^{-\lambda} + \varepsilon_{\mathsf{samp}}(\lambda),$$

where $\varepsilon_{\mathsf{samp}}$ is the joint statistical distance of an implemented Gaussian sampler from the ideal samples and is zero in the ideal distributional model. In the pointwise bound, this joint distribution contains $(s, e)$ and all samples used by the fixed active set. In the uniform bound, it contains $(s, e)$ and the complete preprocessing family for every signer from which the active set may be selected. If only per-sample distances are available, $\varepsilon_{\mathsf{samp}}$ includes their composition over the corresponding batch. A concrete implementation is required to make $\varepsilon_{\mathsf{samp}}(\lambda)$ negligible. This is a pointwise bound for each fixed admissible $\mathsf{act}$ chosen independently of the preprocessing randomness. For active-set selection after observing the public tokens, let

$$S(n, t, a_{\max}) = \sum_{a=t}^{a_{\max}} \binom{n}{a}.$$

For each admissible $\mathsf{act}$, let $(\sigma_{\mathsf{act}}, \mathcal{B}_{\mathsf{act}})$ be the result of an honest counterfactual execution from a separate copy of the same fresh post-preprocessing state, using the common sampled key and token family. Each copy binds only its own context. The event below means that every such choice would succeed if selected; only one choice is executed in the real protocol. A union bound on the Gaussian norm events before choosing the active set gives the uniform event

$$\begin{aligned}
&\Pr\left[\begin{array}{c}\forall \mathsf{act} \subseteq [n], \quad t \leq |\mathsf{act}| \leq a_{\max} :\\ \mathsf{Vf}^{\mathcal{B}_{\mathsf{act}}}(\mathsf{vk}, \mathsf{msg}, \sigma_{\mathsf{act}}) = 1\end{array}\right]\\
&\qquad \geq 1 - 6S(n, t, a_{\max})2^{-\lambda} - \varepsilon_{\mathsf{samp}}(\lambda).
\end{aligned}$$

Thus adaptive post-preprocessing selection uses $\varepsilon_{\mathsf{corr}}^{\mathsf{unif}} \le 6S(n,t,a_{\max})2^{-\lambda}+\varepsilon_{\mathsf{samp}}$. The supported parameter regime must restrict $(n,t,a_{\max},N)$ so that the chosen pointwise or uniform error target is negligible.

## B. Parameter Analysis

a) Inputs and conventions: We use $R_q = \mathbb{Z}_q[X]/(X^N+1)$, centered representatives, and $D_\sigma(x) \propto \exp(-\pi x^2/\sigma^2)$. Every $\sigma$ in this profile, including $\sigma_{\mathsf{td}}$, denotes a Gaussian width. The corresponding standard-deviation parameter is $\sigma/\sqrt{2\pi}$. Fix $n = t = a_{\max} = 1024$, a challenge-query budget $Q = 2^{60}$, and $\epsilon = 2^{-320}$ for checking the smoothing and trapdoor inequalities. The statistical parameter does not itself certify the error of an implemented sampler. The revised prime modulus satisfies $q \equiv 1 \pmod{2N}$.

| Parameter | TT-128-R1 |
|---|---|
| $N$ | 4096 |
| $q$ | 18014398509309953 |
| $\tau$ | 32 |
| $(\sigma_1,\sigma_2,\sigma_r)$ | $(2^6, 2^{40}, 2^{31})$ |
| $\sigma_{\mathsf{td}}$ | 4349014 |
| $(p_1,p_2)$ | $(38,42)$ |
| $(q_1,q_2)$ | $(65536,4096)$ |

Here $q_i = \lfloor q/2^{p_i} \rceil$ exactly. Although $p_1, p_2$ increase, the larger modulus gives larger compressed alphabets and thus less relative compression than the retired profiles.

b) Exact verification thresholds: Use $\alpha = e^{1/5}/\sqrt{2\pi}$ and the preceding correctness formulas to obtain ideal bounds $\beta_z^{\mathrm{ideal}}$, $\beta_r^{\mathrm{ideal}}$, and $\beta_\Delta^{\mathrm{ideal}}$. Round their squares outwards:

$$K_z = \lceil (\beta_z^{\mathrm{ideal}})^2 \rceil, \quad K_r = \lceil (\beta_r^{\mathrm{ideal}})^2 \rceil, \quad K_\Delta = \lceil (\beta_\Delta^{\mathrm{ideal}})^2 \rceil.$$

The exact constants are

$$\begin{aligned} K_z &= 1203919592477717581798617694083, \\ K_r &= 4592588776922465877528417, \\ K_\Delta &= 142861. \end{aligned}$$

The verifier compares the integer squared norms with these integers. For this profile, use the actual radii $\beta_z = \sqrt{K_z}$, $\beta_r = \sqrt{K_r}$, and $\beta_\Delta = \sqrt{K_\Delta}$ throughout extraction. Numerically,

$$\begin{aligned} \beta_z &\simeq 1.097232697506649 \cdot 10^{15}, \\ \beta_r &\simeq 2.143032612192933 \cdot 10^{12}, \\ \beta_\Delta &\simeq 377.9695754952. \end{aligned}$$

Substituting these actual radii into the complete extraction formula gives

$$\begin{aligned} \beta_\delta &\simeq 4.169080475712151 \cdot 10^{15}, \\ \beta_{\mathsf{SIS}} &\simeq 4.711361666570529 \cdot 10^{15} \\ &\simeq 0.2615331099806451q < (q-1)/2. \end{aligned}$$

This removes the previous universal public-column short-solution obstruction; it does not prove normalized ring-SIS hardness.

c) Strict width checks and correctness: We use the strict Hint-RLWE relation in TalonG's Theorem 1, retaining $\tau^2$ rather than replacing the relaxed asymptotic factor by one:

$$\sigma = \left(\frac{2}{\sigma_1^2} + \frac{2Q\tau^2}{\sigma_2^2}\right)^{-1/2} \simeq 20.2385770251.$$

A sufficient upper bound on $\sqrt{2}\eta_\epsilon(\mathbb{Z}^N)$ is $\sqrt{2\ln(2N(1+1/\epsilon))/\pi} \simeq 12.1220175590$. The selected ephemeral and preimage widths satisfy

$$\begin{aligned} &\sigma_2 > 2\sqrt{N}q^{1/2+1/N} \simeq 1.7337581266 \cdot 10^{10}, \\ &\sigma_r > 1.17\sqrt{q}\sqrt{\ln(4N(1+1/\epsilon))/\pi} \simeq 1.3480523948 \cdot 10^9. \end{aligned}$$

The second inequality is sufficient for the cited smoothing bound only when the generated trapdoor satisfies $\|\widetilde{B}\|_{\mathsf{GS}} \le 1.17\sqrt{q}$. Falcon's sampling choice $1.17\sqrt{q/(2N)}$ uses a standard-deviation parameter, whereas this profile uses Gaussian widths [?]. The corresponding width and our rounded candidate are

$$\begin{aligned} \sigma_{\mathsf{td}}^{\mathsf{Falcon}} &= \sqrt{2\pi}\,1.17\sqrt{q/(2N)} \\ &\simeq 4349013.011830253, \\ \sigma_{\mathsf{td}} &= \lceil \sigma_{\mathsf{td}}^{\mathsf{Falcon}} \rceil = 4349014. \end{aligned}$$

This conversion replaces the former width 1735006; it is not a universal trapdoor-width lower bound or a proof that a particular $\mathsf{TDGen}$ is efficient. At the chosen $(N,q)$, the generator must still establish its acceptance probability, Gram–Schmidt bound, and public-key distribution, and $\mathsf{Pre}$ must establish its conditional sampling guarantee. Correcting $\sigma_{\mathsf{td}}$ does not change $\sigma_1, \sigma_2, \sigma_r$ or the displayed correctness and extraction radii. In particular, $\sigma_r$ is not rescaled mechanically. These scalar inequalities and the verification thresholds are numerical checks, not a certification of those trapdoor guarantees. Since $n = t = a_{\max}$, the active-set union factor is one. The six-event ideal correctness tail is at most

$$6\exp\left(N\left[\frac{1}{5} + \frac{1-e^{2/5}}{2}\right]\right) \simeq 2^{-268.72391}$$

per signing execution, plus the joint sampler error for that execution. This is not an experiment-wide sampler-error bound.

d) Estimator results and their scope: We ran the unmodified Lattice Estimator at commits 5ba00f5 (the supplied TalonG reference) and 53da598 (a second version check), using SageMath 10.9, the classical MATZOV model, and the explicit gsa shape model. Gaussian widths are converted to standard deviations by division by $\sqrt{2\pi}$. The LWE and SIS inputs are unchanged by the trapdoor-width correction, so their recorded outputs below are retained. The NTRU run would use the corrected standard-deviation parameter $4349014/\sqrt{2\pi}$, together with an analysis of the screened key distribution defined below. We defer this numerical evaluation. The conditional theorem retains the screened-key indistinguishability assumption; deferring an attack-cost estimate does not discharge it. The generic coefficient-lattice mappings are LWE $(n,m) =$

$(N, N)$, Euclidean SIS $(n, m) = (N, 4N)$, and circulant NTRU $(n, m) = (N, N)$. These mappings are heuristic for the structured ring instances used here, including normalized ring-SIS.

| Problem | MATZOV bits | BKZ block |
|---|---|---|
| Hint-RLWE (BDD) | 301.11 | 956 |
| SIS (full bound) | 299.04 | 949 |
| NTRU (uSVP) | not evaluated | — |

The first two rows are the lowest finite, usable outputs in the recorded MATZOV runs for those problems. With the classical ADPS16 model, LWE uSVP costs $2^{280.904}$ and SIS costs $2^{277.108}$. These are model-specific work estimates, not bounds on distinguishing advantage. The former NTRU estimate of 971.24 bits at block size 3335 used the superseded trapdoor-width input and was also an extrapolation beyond the fitted block-size range. It is withdrawn for this corrected profile. Other recorded NTRU calls encountered model limits or returned non-finite outputs. The recorded BKW call failed numerically. All diagnostics are retained in the reproducibility package; a failed call or infinite output does not certify security or establish the minimum over all attacks.

e) Interpretation of heuristic estimates: The cost models and generic coefficient-lattice mappings above are used as explicitly heuristic evidence about known attacks. They are not claimed as proven lower bounds against all algorithms. Such estimates can guide a candidate parameter choice without a tight concrete security reduction, provided that the instance distribution, attack coverage, cost model, and extrapolations are stated. In contrast, simulation identities, extraction bounds, and statistical-distance composition in the conditional theorem require mathematical justification. An unproved primitive property must be identified as an assumption rather than inferred from a favorable estimator output. In particular, the omitted NTRU estimate leaves this profile without a complete numerical assessment of all assumptions used by the proof.

f) Additional screens and communication: The challenge entropy is $\tau + \log_2 \binom{N}{\tau} \simeq 298.1615854$ bits. The correction radius is strictly below $(q_2/2)\sqrt{N}$. For only the fresh, uniform-$C$, $z = r = 0$ strategy with a fresh challenge-oracle input, the finite-alphabet Chernoff bound is less than $2^{-30272}$ per trial. This does not bound a fixed or adversarially selected $C$, or general forgeries. Simple fixed-width encoding uses 51, 42, and 10 bits per coefficient of $z, r, \Delta$, respectively. Their total is 52,736 bytes (51.5 KiB), excluding the challenge and metadata. Encoding full $a$ and compressed $b$ takes 35 KiB; one uncompressed public token polynomial takes 27 KiB. A canonical sparse challenge representation takes 52 bytes. These are encoding budgets, not measured protocol bandwidth or runtime results.

g) Remaining proof and implementation obligations: The state-assisted theorem eliminates $E_1$ in the ideal service model, but its $E_0$ term still incurs label-guessing and forking losses. No end-to-end 128-bit claim follows from the estimator outputs alone. Such a concrete claim requires explicit experiment-wide budgets, including $L, T$, PRF and sampler errors, supported primitive advantage bounds, and a realization of the trusted service. A trapdoor implementation must supply the required guarantees at this modulus and dimension. Changing the supported participant sets, query budget, or widths requires recomputation of both correctness and security conditions.